



MindHx

Notice the signal. Keep the human.

An early-detection mental-health triage aid that fuses voice, language, and validated clinical questionnaires into one explainable risk signal – private by default, and bilingual in English and Urdu from the ground up.

Next.js 16

FastAPI

Qwen / DashScope

PHQ-9 · GAD-7 · K10

English + اردو

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AGENDA

What we'll *cover*.

- **01** Introduction and why MindHx matters
- **02** Why this matters for Pakistan
- **03** The product – screenshots and functionality
- **04** The AI chatbot – everyday support
- **05** Urdu language – the core of MindHx's accessibility mission
- **06** Technical architecture, quality, and safety
- **07** Roadmap and future possibilities
- **08** Team and closing

Lower the barrier *to a first screening step.*

Depression, anxiety, and related conditions are frequently under-screened, especially where access to mental-health professionals is limited and stigma discourages self-report. MindHx's objective is a private, low-friction check-in that surfaces a risk signal a person can act on – explicit about what it is not: not a diagnosis, not a replacement for a clinician, not a clinically calibrated instrument (yet).

Multimodal, not single-signal

Voice, text, and three validated questionnaires are fused into one explainable score with a genuine per-signal attribution breakdown.

Route, don't diagnose

Risk-tier language throughout, never diagnostic language. Crisis signals short-circuit everything and go straight to Emergency Support.

Minimal data, accounts optional

The core check-in never requires an account. Audio is processed and discarded. Only score/band/themes are ever saved.

01 / INTRODUCTION – WHY MINDHX MATTERS

MindHx matters because mental health screening is still hard and expensive to access, especially in a country like Pakistan where psychiatrists are scarce and social stigma keeps people from speaking up.

It combines three signals at once – how someone sounds, what they say, and validated PHQ-9, GAD-7, K10 questionnaires – into one clear score with a breakdown of what contributed to it.

Rather than a formal diagnosis, it assesses the severity of the situation and underlying risk factors by detecting themes like grief, trauma, suicide risk, anxiety, depression, and pain, and uses this to route elevated risk straight to a professional. If something like self-harm comes up, it redirects to emergency support immediately, before any score is even shown.

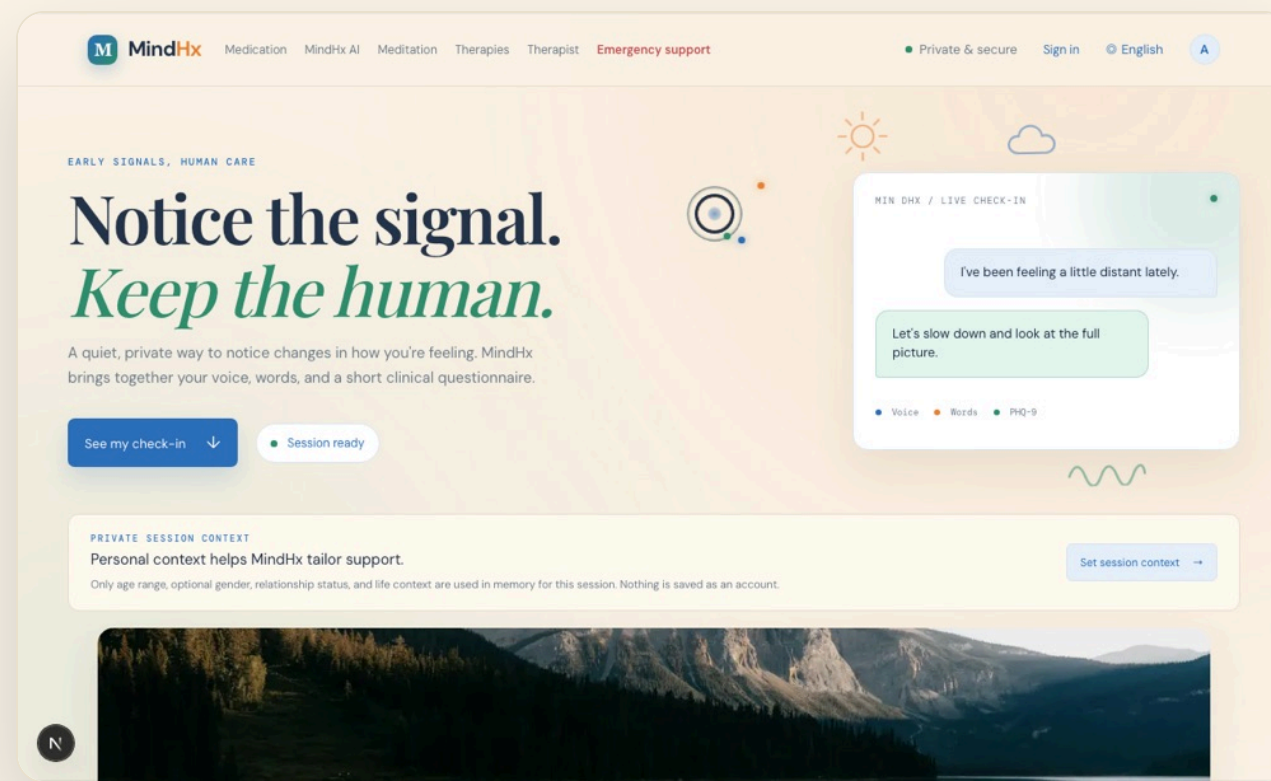
Signing in is optional, only for people who want their check-in history saved, and even then only the score and general themes are stored, never a transcript or written answers. Audio itself is always deleted right after processing. It also works fully in Urdu and English, making it usable locally rather than just another English-only app.

A screening gap that is *structural, not incidental.*

- **Access:** Pakistan has one of the world's lowest psychiatrist-to-population ratios – widely cited estimates put it under 500 psychiatrists for a population of 240+ million, concentrated in a handful of major cities.
- **Stigma:** seeking mental-health help is often read as personal weakness or family shame, especially outside urban centers, which keeps people from taking even a first, private step.
- **Language:** Urdu is the national language and lingua franca, yet most digital mental-health tools are English-only – quietly excluding the people most likely to need them.
- **Cost and distance:** a first appointment often requires travel, time off work, and a fee just to find out whether something is worth pursuing further.

MindHx is built directly against these four barriers: private and free to try, no account required, fully bilingual, and paired with a real, verified directory of Pakistani providers for the moment a person is ready for a real conversation.

Voice, words, and a *short clinical questionnaire* — in one flow.



How the combined score is calculated

Each signal – acoustic, linguistic, and clinical – is scored independently first, then combined into one weighted estimate using an additive model. Each term's share in the final number is the real, mathematically exact contribution, not an approximation bolted on afterward – shown directly on the results page.

3

FUSED SIGNALS

27

PHQ-9 MAX SCORE

21

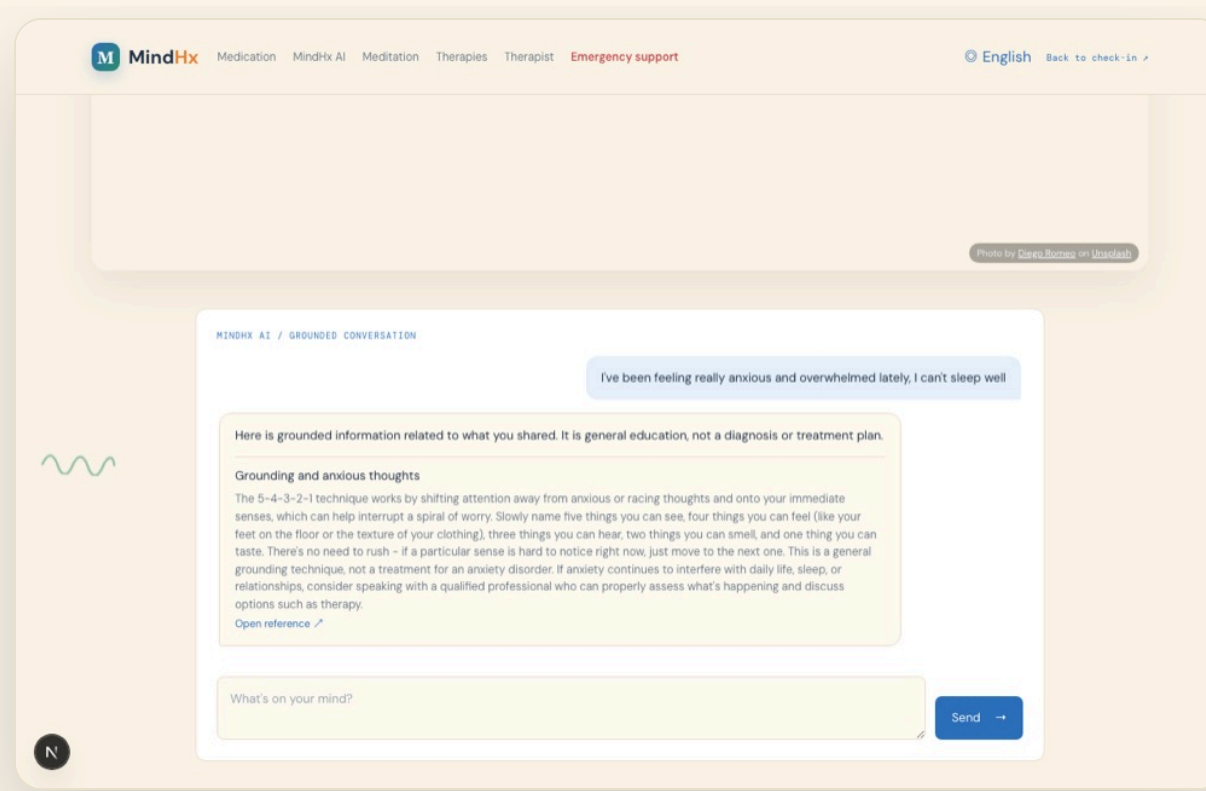
GAD-7 MAX SCORE

50

K10 MAX SCORE

If the voice signal reads flat but the questionnaire answers are mild – or the other way around – that tension is shown honestly, not smoothed into one tidy number.

MindHx AI — *a real grounded reply,* not a canned demo.



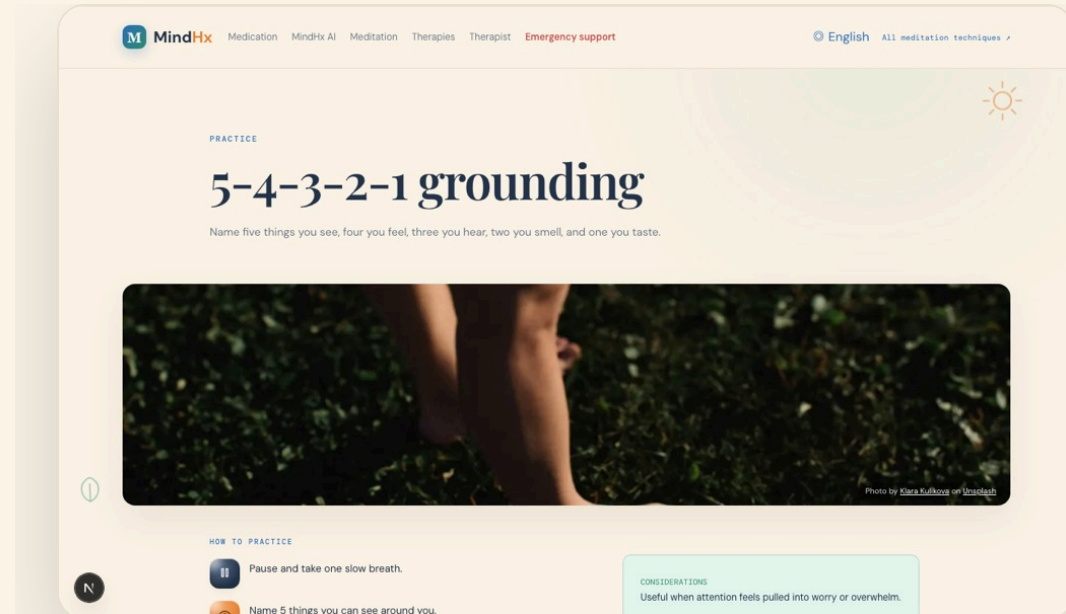
A middle ground between doing nothing and seeing a professional

MindHx also includes an AI chatbot built for everyday stress, anxiety, and burnout – the kind of thing most people never bring up with anyone. It only answers from a fixed, reviewed set of information, so there is no risk of unsafe advice.

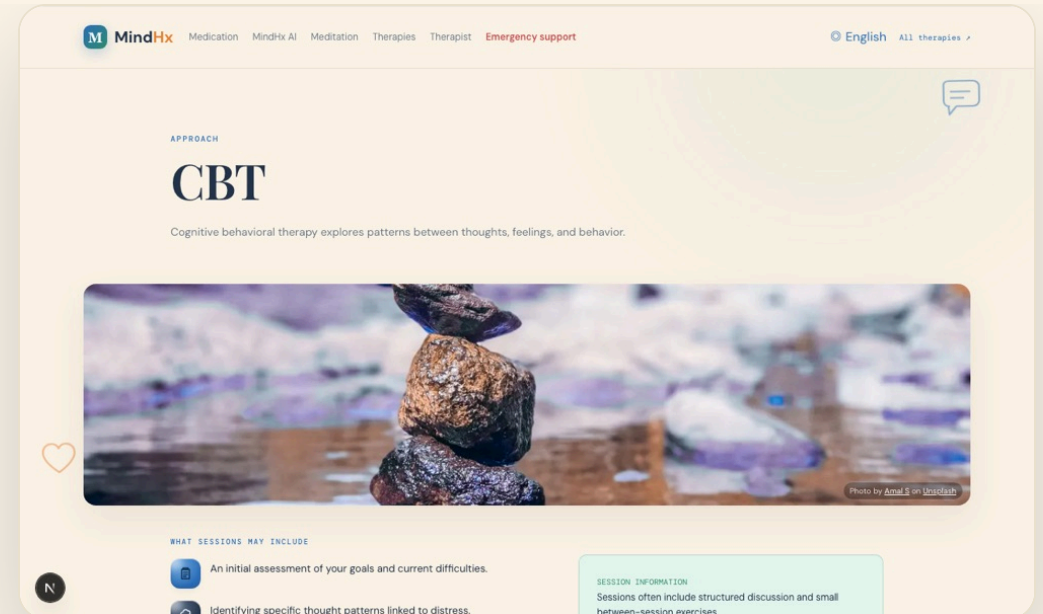
Most people here never see a professional for smaller issues like poor sleep, work pressure, or daily anxiety. Through the chatbot they get simple meditation techniques and basic therapy approaches right away – no appointment, no waiting.

It fills a middle ground. Not a replacement for real treatment, but not leaving someone with nothing either. If the conversation turns serious, it stops and redirects to emergency support instead of continuing.

A practice library with *step-by-step 3D icons.*

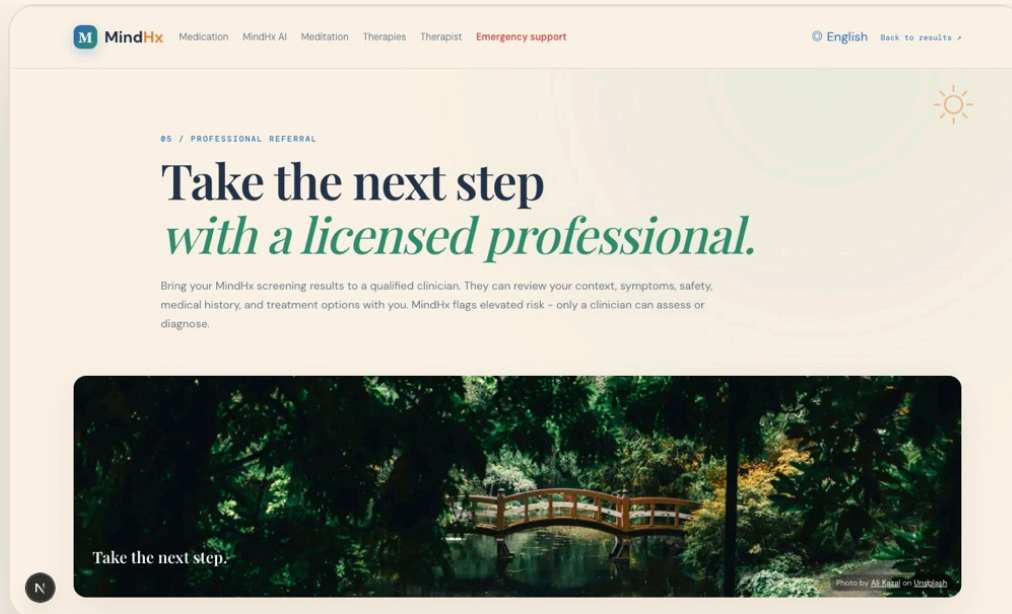


5-4-3-2-1 grounding – original 3D-shaded step icons

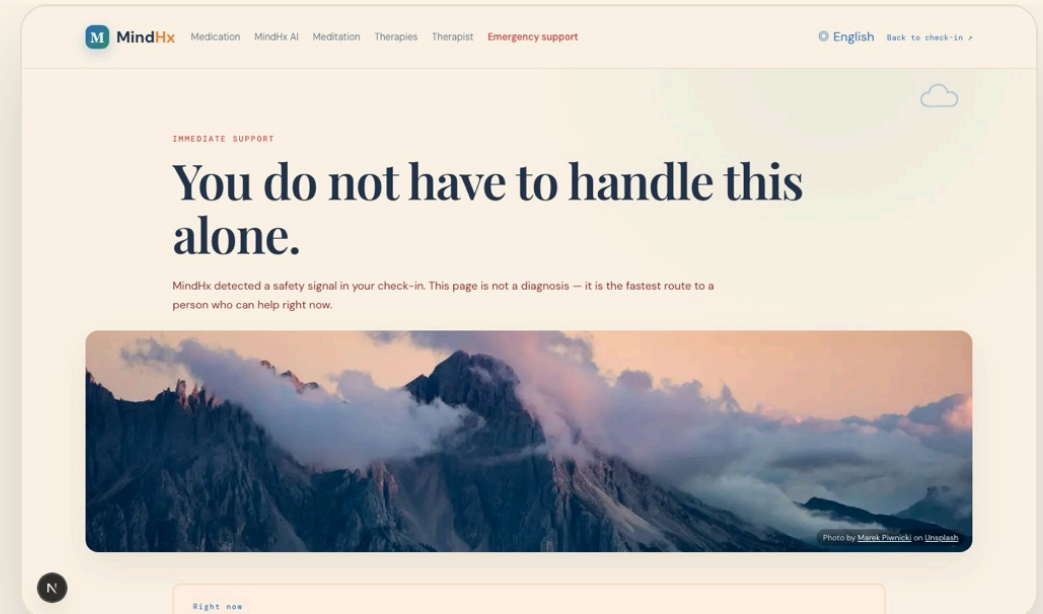


CBT – what sessions may include

From a private check-in *to a real next step.*



Verified Pakistan therapist/psychiatrist directory



Emergency support — reachable at any point in the flow

05 / CORE FOCUS

Urdu Language

*Not a translation layer —
a core requirement.*

The single most important design decision in MindHx: built bilingual from the ground up, not translated as an afterthought.

Why Urdu, and *why it had to be built in,* not bolted on.

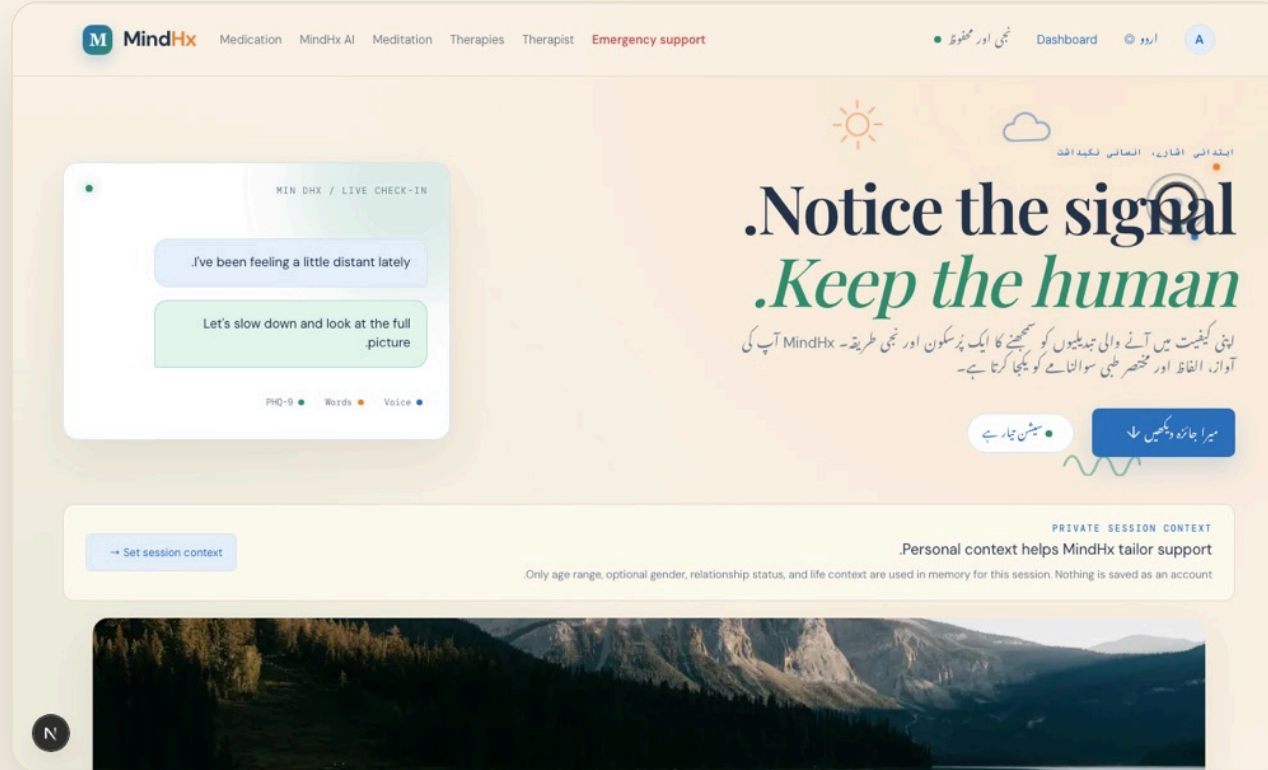
Every page – the check-in itself, the AI chat, the therapist directory – exists in both English and Urdu, switchable with a single tap in the header, with the entire layout correctly mirroring for Urdu's right-to-left script rather than just swapping words inside a left-to-right frame. A screening tool available only in English quietly excludes most of the people who might actually need it in a country where Urdu is the lingua franca.

● Header-level toggle on every page

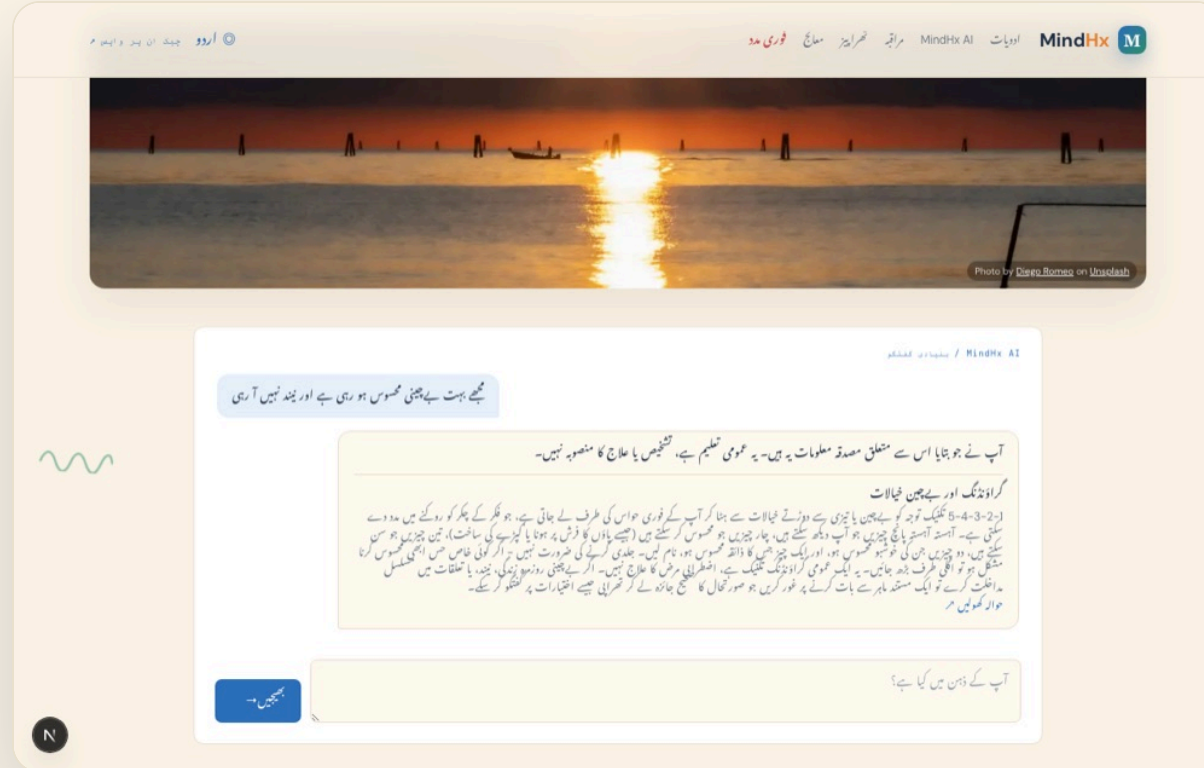
● Full RTL layout mirroring

● Server-side Urdu-script safety guard

آہستہ چلیں۔ اشارے کو محسوس کریں۔



MindHx AI – ایک حقیقی اردو گفتگو



MindHx کے اندر ایک AI چیٹ بوٹ بھی شامل ہے جو روزمرہ کے عام ذہنی دباؤ، پریشانی اور تھکن جیسے مسائل میں فوری رہنمائی فراہم کرتا ہے۔ یہ چیٹ بوٹ آزادانہ طور پر کچھ بھی نہیں بناتا بلکہ صرف پہلے سے طے شدہ اور محفوظ معلومات کی بنیاد پر جواب دیتا ہے، تاکہ کوئی غلط یا خطرناک مشورہ نہ دیا جائے۔ پاکستان میں زیادہ تر لوگ چھوٹے موٹے ذہنی مسائل جیسے نیند کی کمی، کام کا دباؤ، گھریلو ذمہ داریوں کا بوجھ یا روزمرہ کی بے چینی کے لیے کسی ماہر کے پاس نہیں جاتے۔ اس چیٹ بوٹ کے ذریعے ایسے لوگوں کو سادہ مراقبہ کی تکنیکیں اور بنیادی تھراپی کے طریقے فوری طور پر مل جاتے ہیں، بغیر کسی اپائنٹمنٹ یا انتظار کے۔

مثال کے طور پر اگر کسی کو نیند نہیں آرہی یا ذہنی تناؤ محسوس ہو رہا ہے، تو وہ فوراً سانس لینے کی مشقیں، ذہنی سکون کی تکنیکیں یا سادہ طرز عمل تھراپی کے مشورے حاصل کر سکتا ہے۔

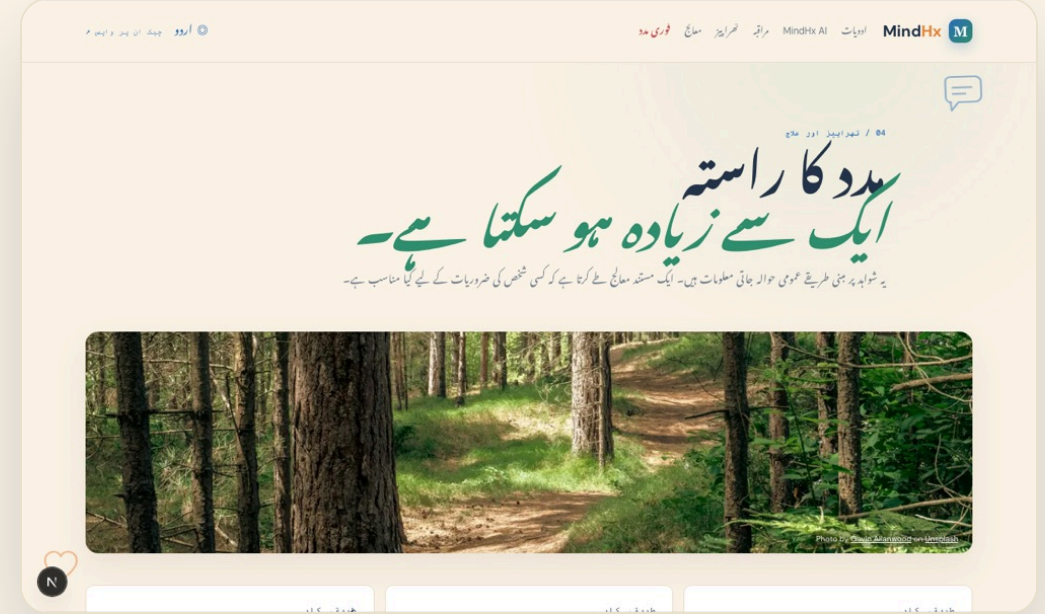
یہ خاص طور پر ان لوگوں کے لیے مفید ہے جو یہ سمجھتے ہیں کہ ان کا مسئلہ اتنا بڑا نہیں کہ ڈاکٹر کے پاس جایا جائے، مگر پھر بھی روزمرہ زندگی متاثر ہو رہی ہوتی ہے۔ اس طرح یہ چیٹ بوٹ ایک درمیانی درجے کی مدد فراہم کرتا ہے۔ نہ تو یہ طبی علاج کا متبادل ہے اور نہ ہی صارف کو بغیر رہنمائی کے چھوڑتا ہے۔

اگر کبھی بات زیادہ سنگین ہو جائے، جیسے خودکشی کے خیالات ظاہر ہوں، تو چیٹ بوٹ فوراً بات چیت روک کر صارف کو ایمرجنسی سپورٹ کی طرف بھیج دیتا ہے۔ اس طرح MindHx صرف ایک سکریٹنگ ٹول نہیں بلکہ روزمرہ کی ذہنی صحت کی دیکھ بھال کا ایک عملی ساتھی بھی بن جاتا ہے، خاص طور پر اس ملک میں جہاں تھراپی تک رسائی محدود اور مہنگی ہے۔

مراقبے کی تکنیکیں اور تھراپیز

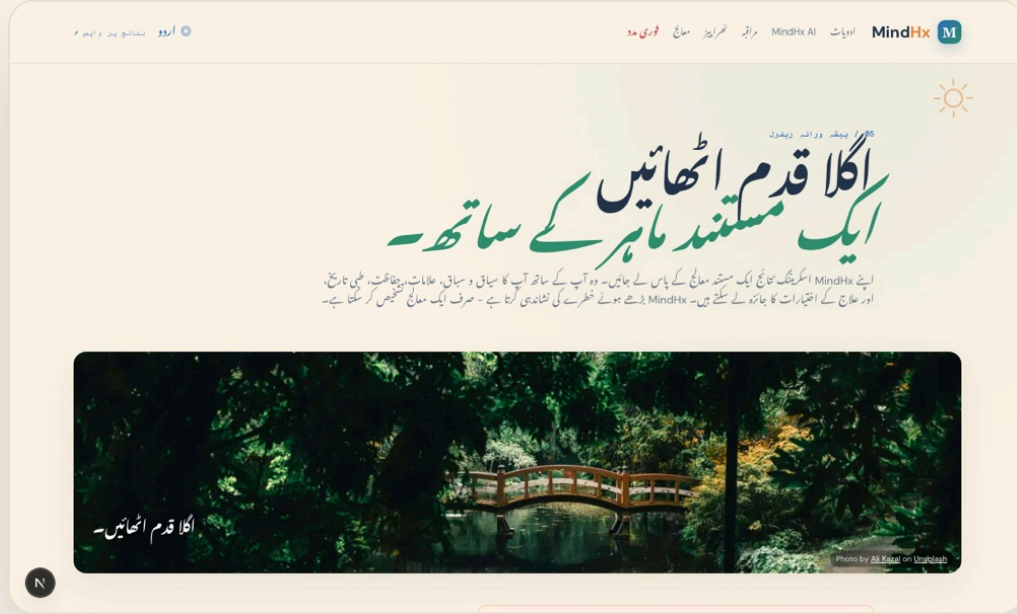


مراقبے کی تکنیکیں - اردو

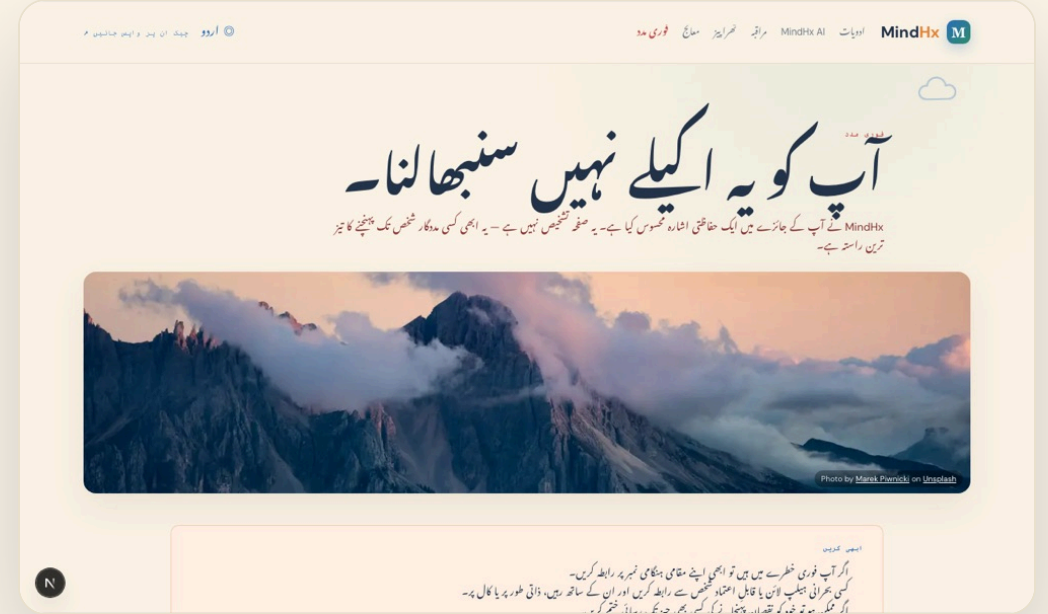


تھراپیز - اردو

معالج تلاش کریں اور فوری مدد



معالج تلاش کریں - اردو



فوری مدد - اردو

Defense in depth: *never show an English reply* when Urdu was requested.

The AI's system prompt instructs Qwen to reply in the requested language – but LLMs don't always comply. MindHx adds a server-side guard, `is_urdu_script()`, that checks the letter composition of any generated reply. If Urdu was requested and the reply reads as English, it is discarded and replaced with the guaranteed-correct deterministic Urdu template – the same defense-in-depth pattern already used for crisis-language re-checking.

Backend, main.py

```
if language == "ur" and not is_urdu_script(result["message"]):  
    return None # fall back to the verified-Urdu template
```

A careful draft, *not a validated instrument* — and we say so.

The Urdu translation of PHQ-9, GAD-7, and K10 is a careful draft for this session, not a licensed clinical instrument – stated plainly on-screen whenever Urdu is active, not hidden in fine print. Item 9 (self-harm ideation) is safety-critical in the validated instrument; shipping an unlicensed translation as clinical-grade would be irresponsible.

The roadmap is explicit about this: partnering for a properly validated Urdu instrument is the single highest-leverage localization gap remaining – it would let MindHx move from a bilingual demo to a real screening tool usable in Urdu-speaking clinical settings.

Everything the app *actually does.*

- Voice, text, and PHQ-9/GAD-7/K10 check-in with a live combined-signal preview
- Explainable per-signal attribution on the results page
- Crisis short-circuiting (PHQ-9 item 9, self-harm language) straight to Emergency Support
- Bounded, source-grounded AI chat with interactive exercises (4-7-8 breathing, 5-4-3-2-1 grounding)
- Meditation and therapy reference libraries with original 3D step icons
- Verified Pakistan therapist/psychiatrist directory, city by city
- Full English/Urdu bilingual UI with a header-level toggle, everywhere
- Optional accounts (bcrypt + JWT) for saved check-in history – score/band/themes only

Two services.

Stateless by default.

A Next.js 16 (App Router) frontend and a FastAPI backend. The screening/risk-assessment endpoints are fully stateless – every request self-contained, nothing persisted. An optional layer adds PostgreSQL-backed accounts purely for people who choose to save history.

20

BACKEND ENDPOINTS

12

FRONTEND ROUTES

23

BACKEND TESTS

2

LANGUAGES

Selected backend endpoints

Endpoint	Purpose
POST /analyze-voice	Heuristic prosodic signal (pause ratio, loudness variability, speaking rate) — pluggable via VOICE_BIOMARKER_PROVIDER
POST /analyze-text	Sentiment + crisis-language detection — Qwen (DashScope) → OpenRouter → local heuristic fallback
POST /ai/chat	Bounded, source-grounded chat; safety-gated before any content is returned; server-side Urdu-script guard on generated replies
POST /risk-assess	Fuses all signals into the combined score, band, explanation, and routing decision
POST /score-phq9, /score-gad7, /score-k10	Individual validated-questionnaire scoring
POST /auth/register, /auth/login	Optional account creation/sign-in — bcrypt + JWT
POST /synthesize	Urdu text-to-speech via Uplift AI

Verified, not *assumed.*

23 backend tests

Crisis short-circuiting, bilingual AI chat responses, trend-aware replies, risk-fusion math, and the full account flow.

Type-checked & linted

tsc --noEmit and eslint run clean on every change – no silent runtime surprises.

Real screenshot QA

Every visual change verified via headless-Chrome screenshots in both English and Urdu, not assumed from the code.

Defense in depth

Crisis-language re-check and the Urdu-script guard both discard unsafe generated output rather than trusting the model once.

What's built — *and what's deliberately not.*

BUILT

- Voice + text + PHQ-9/GAD-7/K10 fused into one explainable score
- Bounded AI chat with a hard safety gate, in English and Urdu
- Original 3D step-icon library across every technique/therapy page
- Real Pakistan therapist/psychiatrist directory
- Full site SEO (per-page metadata, sitemap, structured data)

NOT YET BUILT — DELIBERATELY

- **Score calibration** — no labeled outcome data yet
- **Clinically validated Urdu PHQ-9/GAD-7/K10** — current translation is demo-only
- **Real acoustic biomarker vendor** — today's signal is a heuristic proxy
- **Auth hardening** — rate-limiting and httpOnly cookies still pending

Where this goes *from here.*

- **A validated Urdu instrument** – partnering to properly validate the PHQ-9/GAD-7/K10 Urdu translation would unlock real clinical deployment in Urdu-speaking settings, not just a bilingual demo.
- **Score calibration** – once labeled outcome data exists, Platt scaling or isotonic regression would let the score become an actual probability.
- **A real acoustic biomarker vendor** – upgrading the voice signal from a heuristic proxy to a validated one.
- **Deeper Alibaba Cloud integration** – Qwen/DashScope is already live; Function Compute and OSS-backed caching are natural next steps.
- **More regional languages** – the same bilingual architecture (per-page copy objects, RTL-aware layout) generalizes to Punjabi, Pashto, or Sindhi.

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Keep the human.

MindHx is a screening and triage aid, not a diagnosis. Every score is a starting point for a conversation with a qualified professional — never a replacement for one.